ACADEMIC CENTRAL

PRACTICE EXAMINATION

NC(V) LEVEL 2

MATHEMATICS

PRACTICE EXAM · MARKING GUIDELINE

MARKS: 100

Marking guideline for the Academic Central NC(V) Level 2 Mathematics practice exam.

Written by Academic Central for practice, following the NC(V) subject guidelines for NC(V) Level 2. It is not a Department of Higher Education and Training marking guideline.

How to mark: each ✓ is one mark. Give method marks for correct working even if the final answer is wrong (consistent accuracy). Accept other correct methods.

QUESTION 1

1.1.1 $9x^4y^6$ ✓ $\div 9x^3y$ ✓ $= xy^5$ ✓ (3)

1.1.2 $\frac{2^x(2^2-1)}{3\cdot 2^x}$ ✓ $= \frac{2^x\cdot 3}{3\cdot 2^x}$ ✓ $= 1$ ✓ (3)

1.2 $4{,}56\times 10^{-4}$ ✓ (1)

1.3.1 $3x(2x-5)$ ✓ (1)

1.3.2 $(x-3)(x-4)$ ✓✓ (2)

1.3.3 $(2x-5)(2x+5)$ ✓✓ (2)

1.4 $\frac{(x-3)(x+3)}{(x+3)(x-2)}$ ✓✓ $= \frac{x-3}{x-2}$ ✓ (3)

1.5.1 $3x-6=2x+5$ ✓; $x=11$ ✓ (2)

1.5.2 $(x-7)(x+2)=0$ ✓; $x=7$ or $x=-2$ ✓ (2)

1.5.3 $-2x>6$ ✓; $x<-3$ ✓ (sign reversed) (2)

1.6 Adding: $3x=12$, $x=4$ ✓; $y=3$ ✓ (2)

1.7 $r^2=\frac{V}{\pi h}$ ✓; $r=\sqrt{(\frac{V}{\pi h})}$ ✓ (2)

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QUESTION 2

2.1.1 y-intercept: $(0\,;-4)$ ✓; x-intercept: $(2\,;0)$ ✓ (2)

2.1.2 y-intercept $(0\,;4)$ ✓; x-intercepts $(-2\,;0)$ and $(2\,;0)$ ✓; turning point $(0\,;4)$ ✓ (3)

2.1.3 Straight line through (0 ; −4) and (2 ; 0) ✓✓; parabola opening downwards (shape) through (−2 ; 0), (2 ; 0) with turning point (0 ; 4) ✓✓ (4)

2.1.4 $2x-4=-x^2+4$ ✓; $x^2+2x-8=0$; $(x+4)(x-2)=0$ ✓; points $(2\,;0)$ ✓ and $(-4\,;-12)$ ✓ (4)

2.1.5 $-4\le x\le 2$ ✓✓ (2)

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QUESTION 3

3.1.1 $A=P(1+in)=12\,000(1+0{,}08\times 3)$ ✓✓ $=R14\,880$ ✓ (3)

3.1.2 $A = P(1 + i)^n = 12\,000(1{,}08)^3$ ✓✓ = R15 116,54 ✓ (3)

3.1.3 15 116,54 − 14 880 = R236,54 ✓ (1)

3.2 Deposit = R600; balance = R5 400 ✓; interest = 5 400 × 0,15 × 2 = R1 620 ✓; total = R7 020 ✓; instalment = 7 020 ÷ 24 ✓ = R292,50 ✓ (5)

3.3 $20(1{,}06)^2$ ✓ = R22,47 ✓ (2)

3.4 250 × 18,50 = R4 625 ✓ (1)

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QUESTION 4

4.1.1 $AC^2 = 8^2 + 15^2 = 289$ ✓; AC = 17 cm ✓ (2)

4.1.2 $\sin A = {}^{15}/_{17}$ ✓; $\cos A = {}^{8}/_{17}$ ✓; $\tan A = {}^{15}/_{8}$ ✓ (3)

4.1.3 $\tan A = {}^{15}/_{8}$ ✓; $\hat{A} = 61{,}93°$ ✓ (2)

4.2.1 $\sin 70° = {}^{h<}/_{i>/6}$ ✓; *h = 5,64 m* ✓ *(2)*

4.2.2 $\cos 70° = {}^{d<}/_{i>/6}$ ✓; *d = 2,05 m* ✓ *(2)*

4.3.1 *r = 0,7 m* ✓; $V = \pi(0{,}7)^2(2)$ ✓ = $3{,}08\ m^3$ ✓ *(3)*

4.3.2 *3,0788 × 1 000 ≈ 3 078,76 ℓ* ✓ *(1)*

4.3.3 $SA = 2\pi r^2 + 2\pi rh$ ✓ = $2\pi(0{,}7)^2 + 2\pi(0{,}7)(2)$ ✓ = $11{,}88\ m^2$ ✓ *(3)*

4.4.1 *Floor = 4,5 × 3,2 = 14,4* m^2 ✓; *tile = 0,4 × 0,4 = 0,16* m^2 ✓; *14,4 ÷ 0,16 = 90 tiles* ✓ *(3)*

4.4.2 *90 ÷ 12 = 7,5 → 8 boxes* ✓ *(1)*

4.5 *(a) 3,5 m* ✓ *(b) 24 000* cm^2 ✓ *(c) 90 ÷ 3,6 = 25 m/s* ✓ *(3)*

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QUESTION 5

5.1.1 *39; 45; 55; 58; 62; 62; 62; 66; 71; 80* ✓ *(1)*

5.1.2 *Mean = 600 ÷ 10* ✓ *= 60* ✓ *(2)*

5.1.3 *Median = (62 + 62) ÷ 2 = 62* ✓; *mode = 62* ✓ *(2)*

5.1.4 *80 − 39 = 41* ✓ *(1)*

5.1.5 *The mean (or median) ✓ – there are no extreme outliers, so the mean uses all the data / the median is not affected by the low score of 39 ✓ (accept either with a sensible reason)* *(2)*

5.1.6 *6 out of 10 scored 60 or more (62, 62, 62, 66, 71, 80) ✓ = 60% ✓* *(2)*

5.2.1 $^{3}/_{10}$ ✓ *(1)*

5.2.2 $^{8}/_{10} = {}^{4}/_{5}$ ✓✓ *(2)*

5.2.3 0 ✓ *(1)*

5.3.1 *Heading ✓; labelled axes ✓; equal-width bars with gaps ✓; correct heights (18, 8, 10, 4) ✓* *(4)*

5.3.2 $^{18}/_{40} \times 360° ✓ = 162° ✓$ *(2)*

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TOTAL: 100

END OF MARKING GUIDELINE